FBISE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: 2HA-I 2310 — B(4104) / D(8101)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

Note: BIOLOGY HSSC-II. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 4104. Fill the relevant bubble against each question according to curriculum:

Q1. A snail was moving on a table. When the table was tapped snail stopped moving. When tapping was done repeatedly, snail stopped responding to it. Identify the type of behavior shown.

(A) Imprinting
(B) Habituation
(C) Innate
(D) Instinct

Q2. Due to lightning, a forest was set on fire and destroyed. The type of succession which will occur then is termed as:

(A) Xerarch
(B) Secondary
(C) Hydrarch
(D) Primary

Q3. Which one is a steroid hormone?

(A) Adrenaline
(B) Parathormone
(C) Insulin
(D) Aldosterone

Q4. The following diagram shows nephron. Identify the processes which occur at X and Y. (Note: a nephron diagram with X at the glomerulus and Y at the proximal tubule is given.)

(A) Selective Reabsorption, Ultrafiltration
(B) Diffusion, Tubular secretion
(C) Ultrafiltration, Selective Reabsorption
(D) Active Transport, Selective Reabsorption

Q5. Which of the following steps are involved in inhalation? I. Diaphragm contracts; II. Rib cage lowers; III. External intercostal muscles contract; IV. Pressure in lungs decreases.

(A) II, III, IV
(B) I, III, IV
(C) I, II, IV
(D) I, II, III

Q6. The workers in a honey bee hive exhibit:

(A) Aggregation
(B) Territoriality
(C) Altruism
(D) Dominance Hierarchy

Q7. In the given figure of action potential in a Neuron, hyperpolarization is represented by: (Note: a graph of membrane potential vs Time (ms) is given with P at resting potential, R on the rising phase, S at the peak and Q at the undershoot.)

(A) Q
(B) R
(C) S
(D) P

Q8. Which one of the following is NOT a Sexually Transmitted Disease?

(A) Gonorrhea
(B) Syphilis
(C) AIDS
(D) Measles

Q9. The extra embryonic membrane which provides a fluid filled environment for developing fetus is:

(A) Chorion
(B) Allantois
(C) Yolk sac
(D) Amnion

Q10. In pea plant, round and wrinkled seed shape is controlled by R and r alleles respectively. The cross which will give rise 1:1 ratio of both phenotypes can be:

(A) Rr x Rr
(B) Rr x RR
(C) Rr x rr
(D) RR x rr

Q11. Baldness is more common in males than in females because it is a:

(A) Sex influenced trait
(B) Sex limited trait
(C) Y-linked trait
(D) Sex linked recessive trait

Q12. In DNA replication, the primer:

(A) Consists of few DNA nucleotides
(B) Helps to unwind DNA double strand
(C) Prevents DNA strands to pair up
(D) Is a short segment of RNA nucleotides

Q13. After successful implantation of blastocyst into endometrium of uterus, level of which hormone increases?

(A) Human chorionic gonadotrophin
(B) Luteinizing hormone
(C) Oxytocin
(D) Progesterone

Q14. Normally vaccination can be done at any age for:

(A) Measles
(B) Hepatitis
(C) Diphtheria
(D) Polio

Q15. According to Hardy Weinberg Theorem, which one of the following factor cannot change allelic frequency?

(A) Mutation
(B) Random mating
(C) Selection
(D) Gene flow

Q16. Restriction endonucleases can be used in Genetic Engineering to:

(A) Produce cDNA
(B) Cut plasmid DNA
(C) Join DNA at sticky ends
(D) Clone DNA

Q17. In Nitrogen cycle, conversion of Nitrates into nitrogen is termed as:

(A) Ammonification
(B) Assimilation
(C) Denitrification
(D) Nitrification

> Note: —2HA-I 2310-4104—. Page 1 of 2. Page 2 of 2.

> Note: BIOLOGY HSSC–II. Time allowed: 2:35 Hours. Total Marks Sections B and C: 68. NOTE: Answer any fourteen parts from Section 'B' and any two questions from Section 'C'. Write your answers neatly and legibly.

Q. SECTION – B (Marks 42)

Q. 2 Attempt any FOURTEEN parts. All parts carry equal marks. (14 x 3 = 42)

(i) Compare the mechanism of osmoregulation in Marine and Fresh water animals.

(ii) How are 'Counter Current Multiplier' mechanism and 'hormones' involved in concentrating urine?

(iii) The diagram shows structure of a Sarcomere. a. Label the parts A, B, C, and D. b. Mention the changes in C and D in a contracted sarcomere. (Note: a sarcomere diagram with parts labelled A, B, C and D is given.)

(iv) Enlist the cells associated with bones. Also write their respective functions.

(v) Differentiate between two types of Neurotransmitters. Also give one example of each.

(vi) Limbic system lies under cerebrum. Write down its parts and their functions.

(vii) A man and a woman with blood groups A and B respectively have four children all with different blood groups of ABO system. By a cross show genotypes of both parents and offsprings.

(viii) XO-XX is a pattern of sex determination. Explain with example.

(ix) Write about the causes and symptoms of Phenyl Ketonuria.

(x) An experiment is shown in the diagram. a. Who performed this experiment and why? b. Write down the result of experiments A-D. (Note: a diagram of the experiment with Rough Strain (II-R), Smooth Strain (III-S), Heat-killed Smooth Strain, and Rough Strain + Heat-Killed Smooth Strain injected into mice A, B, C, D is given.)

(xi) Briefly explain any three characteristics of Genetic Code.

(xii) What is Speciation? Briefly explain Sympatric Speciation?

(xiii) What is Divergent Evolution? Explain with a suitable example.

(xiv) How does Endosymbiont hypothesis explain evolution of Eukaryotes from Prokaryotes?

(xv) Population have many characteristics. Write about any three characteristics.

(xvi) Enlist the sources of Choroflouro Carbons (CFCs). How do they cause Ozone Depletion?

(xvii) a. What is the role of CFTR gene in a normal person? b. Write down symptoms of Cystic Fibrosis. c. Write mechanism of Gene Therapy used to cure Cystic Fibrosis.

(xviii) Elaborate the role of Microbes in Food Processing by giving any three examples.

(xix) Write about two major techniques of animal cell culture.

(xx) What are Acclimatization and Selection? How do they help to improve crops?

Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. Describe stages of Menstrual Cycle in Human Female. Also draw the diagram outline. b. Explain Latent Learning with example.

Q. 4 a. Write the names and functions of hormones produced by Thyroid Gland. Write about the problems related with abnormal secretion of hormones in different stages. b. Explain events of Gastrulation in Human. Also draw the diagram.

Q. 5 a. Describe the mechanism of DNA Analysis using Restriction Fragment Length Polymorphism (RRLP). b. How is Carbon Dioxide transported through blood as Bicarbonate ions? Explain in detail.

> Note: —2HA-I 2310—

VERSION D

> Note: BIOLOGY HSSC-II. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre

> Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 8101. Fill the relevant bubble against each question according to curriculum:

Q1. Which of the following events occur during exhalation? I. Diaphragm relaxes; II. Rib cage lowers; III. External intercostal muscles contract; IV. Pressure in lungs Increases.

(A) I, II, III
(B) II, III, IV
(C) I, III, IV
(D) I, II, IV

Q2. If body is overheated, the body temperature can be regulated by:

(A) More sweating, vasoconstriction
(B) Less sweating, vasodilation
(C) Less sweating, vasoconstriction
(D) More sweating, vasodilation

Q3. Which part of Limbic system is involved in long term memory formation?

(A) Cerebellum
(B) Hippocampus
(C) Amygdala
(D) Hypothalamus

Q4. The type of Agonistic behavior exhibited by mountain gorilla is:

(A) Dominance hierarchy
(B) Altruism
(C) Territoriality
(D) Instinct

Q5. Which one is a Sexually Transmitted Disease?

(A) Syphilis
(B) Measles
(C) Polio
(D) Tetanus

Q6. Leydig cells found in testes are involved in secretion of:

(A) Inhibin
(B) Testosterone
(C) Luteinizing hormone
(D) Nutrients

Q7. All of the following are examples of Reflex Action EXCEPT:

(A) Blinking of eyelids in bright light
(B) Pulling hand away from hot object
(C) Sneezing on tickling nose
(D) Picking pen to write name

Q8. When true breeding red and white flowering plants were crossed, all pink offsprings were produced. When these pink flowered plants were self crossed, the percent of pink flowering offsprings would be:

(A) 25%
(B) 50%
(C) 75%
(D) 100%

Q9. During translation, the amino acyl t-RNA complex attaches to small ribosomal unit at:

(A) P site
(B) E site
(C) mRNA
(D) A site

Q10. Hemophilia is more common in males than in females because this trait is:

(A) Y-linked recessive
(B) Sex influenced
(C) Sex limited
(D) Sex linked recessive

Q11. A female having 44 + XO chromosome, webbed neck and large number of moles is suffering from:

(A) Turner syndrome
(B) Down syndrome
(C) Klinefelter syndrome
(D) Sickle cell anaemia

Q12. The ability of a particular environment that can support maximum individuals is termed as:

(A) Uniform distribution
(B) Clumped distribution
(C) Carrying capacity
(D) Density

Q13. All of the following are Stop Codons EXCEPT:

(A) UGA
(B) UAG
(C) UAA
(D) AUG

Q14. Identify the part where fertilization of ovum occurs. (Note: a diagram of the female reproductive system is given with parts labelled A (ovary), B (fallopian tube), C (uterus) and D (cervix).)

(A) A
(B) B
(C) C
(D) D

Q15. To prove DNA replication is semi conservative, Meselson and Stahl labelled DNA with N^{15}, DNA sample appeared heaviest which was collected after.

(A) 0 minutes
(B) 20 minutes
(C) 40 minutes
(D) 60 minutes

Q16. Which type of behaviour helps animals with short life span to live successfully?

(A) Learning
(B) Innate
(C) Imprinting
(D) Altruism

Q17. All are characteristics of Haemoglobin EXCEPT:

(A) Has more affinity with oxygen
(B) Loses oxygen at PO_2 60mmHg
(C) Transports oxygen
(D) Has four iron atoms in molecule

> Note: —2SA-I 2310-8101 HA—. Page 1 of 2. Page 2 of 2.

> Note: BIOLOGY HSSC–II. Time allowed: 2:35 Hours. Total Marks Sections B and C: 68. NOTE: Answer any fourteen parts from Section 'B' and any two questions from Section 'C'. Write your answers neatly and legibly.

Q. SECTION – B (Marks 42)

Q. 2 Attempt any FOURTEEN parts. All parts carry equal marks. (14 x 3 = 42)

(i) Diagram shows human respiratory system. a. Identify the labelled parts A, B, C and D. b. Write functions of parts C and D. (Note: a diagram of the human respiratory system with parts labelled A (top of trachea), B (trachea), C (bronchiole) and D (diaphragm) is given.)

(ii) a. Write down the two specific symptoms of Sinusitis. b. Enlist divisions of Autonomic Nervous System and write their functions.

(iii) Write about the role of Na^+ and K^+ ions in maintaining Resting Membrane Potential.

(iv) Brifely describe any three problems leading to Female Infertility.

(v) What is Habituation? Explain with the example of squirrel.

(vi) Write about the methods of Regulation of Gene Expression.

(vii) What is meant by Non-Conventional energy source? Enlist any three Non-Conventional Energy sources. What is their advantage to use?

(viii) Write names of the layers present in Uterine wall. Also write their structure and functions.

(ix) Elaborate the role of Foetal Hormones in birth process.

(x) The blood groups of father and mother are AB and O respectively. Write genotypes of parents and offspring by making a cross.

(xi) XY-XX is a pattern of Sex Determination. Explain this pattern by giving example.

(xii) Write about any three factors which can change Allelic Frequency of a population.

(xiii) What is Convergent Evolution? Explain with an example.

(xiv) a. Which theory of evolution is depicted in figure and who proposed it? b. Enlist the main points of this theory? (Note: a diagram of giraffe neck evolution is given.)

(xv) Briefly explain different methods of Nitrogen Fixation.

(xvi) Enlist any three causes and effects of Acid Rain.

(xvii) Which properties should a vector have in it? Why it is used in Genetic Engineering.

(xviii) How does Vaccination help in controlling Hepatitis?

(xix) Write role of Hybridization and Back Cross to improve crops.

(xx) Briefly explain steps of Polymerase Chain Reaction.

Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. Write names and functions of hormones released by Anterior Pituitary Gland which affect growth and metabolism. Also explain problems related to their abnormal secretion. b. Ammonia and urea are excretory products. How is their excretion is related to habitat? Also give examples.

Q. 4 a. Why is Sanger's method of DNA Sequencing called Dideoxy Method? Explain its procedure. b. In the given diagram: (i) Label the parts A, B, C, D, and E. Also write their functions; (ii) How is part D made? What is its role? (iii) Which type of blood vessels are present in part B? (Note: a diagram with parts labelled A, B, C, D and E is given.)

Q. 5 a. The transcribed mRNA in eukaryotes is changed and modified before translation. Explain how it occurs. Also give reasons for the changes. b. Describe the steps of Bone Repair after Simple Fracture.

> Note: —2HA-I 2310 HA—